THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · HARD LEVEL CHAPTER TEST SET 1

Chapter 3: **Pair of Linear Equations in Two Variables** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The pair $kx + 2y = k - 2$ and $8x + ky = k$ has no solution when $k =$ 1

(a) 8 **(b)** -4 **(c)** 4 **(d)** 2

2. The pair $kx + y = k - 1$ and $4x + ky = k$ has infinitely many solutions when $k =$ 1

(a) 1 **(b)** 2 **(c)** 4 **(d)** -2

3. If $11x + 4y = 53$ and $5x + 3y = 30$, then $x + y =$ 1

(a) 8 **(b)** 3 **(c)** 5 **(d)** 15

4. The area (in sq. units) of the triangle formed by the line $2x + 3y = 12$ and the coordinate axes is: 1

(a) 3 **(b)** 12 **(c)** 72 **(d)** 24

5. If $x + y = 7$ and $xy = 10$, then $x^2 + y^2 =$ 1

(a) 49 **(b)** 69 **(c)** 29 **(d)** 39

6. If $5x + 3y = 47$ and $3x + 5y = 41$, then $x - y =$ 1

(a) 3 **(b)** 44 **(c)** 6 **(d)** $\frac{3}{4}$

7. The pair $kx + 3y = k - 3$ and $12x + ky = k$ has no solution when $k =$ 1

(a) 6 **(b)** -6 **(c)** 3 **(d)** 12

8. The pair $kx + 7y = k - 7$ and $28x + ky = k$ has infinitely many solutions when $k =$ 1

(a) 14 **(b)** -14 **(c)** 7 **(d)** 28

9. **Assertion (A):** The graphs of $x + y = 4$ and $x - y = 2$ intersect at the point $(3, 1)$. 1

Reason (R): The coordinates of the point of intersection of the lines of a pair of linear equations give the solution of the pair.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** The pair $x + 2y = 4$, $2x + 4y = 12$ has no solution. 1

Reason (R): Two lines with $\frac{a_1}{a_2} = \frac{b_1}{b_2}$ always intersect at exactly one point.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Find k so that the pair $kx + 2y = k - 2$, $8x + ky = k$ has no solution. 2

12. Solve: $5x + 3y = 47$ and $3x + 5y = 41$. 2

13. Find the points where the line $3x + 2y = 12$ meets the coordinate axes and the area of the triangle so formed. 2

14. Find k so that the pair $kx + 3y = k - 3$, $12x + ky = k$ has no solution. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Solve: $\frac{2}{x} + \frac{3}{y} = 2$ and $\frac{3}{x} + \frac{5}{y} = \frac{19}{6}$. 3

16. Find a and b for which the pair $3x + y = 7$ and $(a + b)x + (a - b)y = 14$ has infinitely many solutions. 3

17. Find the area of the triangle formed by the lines $x + y = 6$, $x - y = 2$ and the x-axis. 3

OR

Solve: $\frac{3}{x} + \frac{4}{y} = \frac{11}{4}$ and $\frac{5}{x} + \frac{2}{y} = \frac{9}{4}$. 3

18. Find a and b for which the pair $5x + 3y = 4$ and $(a + b)x + (a - b)y = 12$ has infinitely many solutions. 3

OR

Find the area of the triangle formed by the lines $2x + y = 8$, $x + y = 5$ and the x-axis. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Rohan buys 3 notebooks and 2 pens for ₹ 70, while Priya buys 2 notebooks and 3 pens for ₹ 65 from the same shop. Let the price of a notebook be ₹ n and of a pen be ₹ p. 5

(i) Write the pair of linear equations.

(ii) Solve by elimination to find the price of a notebook.

(iii) Find the price of a pen.

(iv) Aman has ₹ 200. He buys 6 notebooks and spends the rest on as many pens as possible. How many pens does he buy and how much money is left?

20. A boat goes 21 km upstream and 33 km downstream in 10 hours, and 12 km upstream and 55 km downstream in 9 hours. Find the speed of the boat in still water and the speed of the stream. 5

OR

A man travels 900 km partly by train and partly by bus. If he travels 480 km by train and the rest by bus, he takes $\frac{51}{5}$ hours; if he travels 600 km by train and the rest by bus, he takes $\frac{21}{2}$ hours. Find the speeds of the train and the bus. 5

END OF QUESTION PAPER — **Total: 40 Marks**